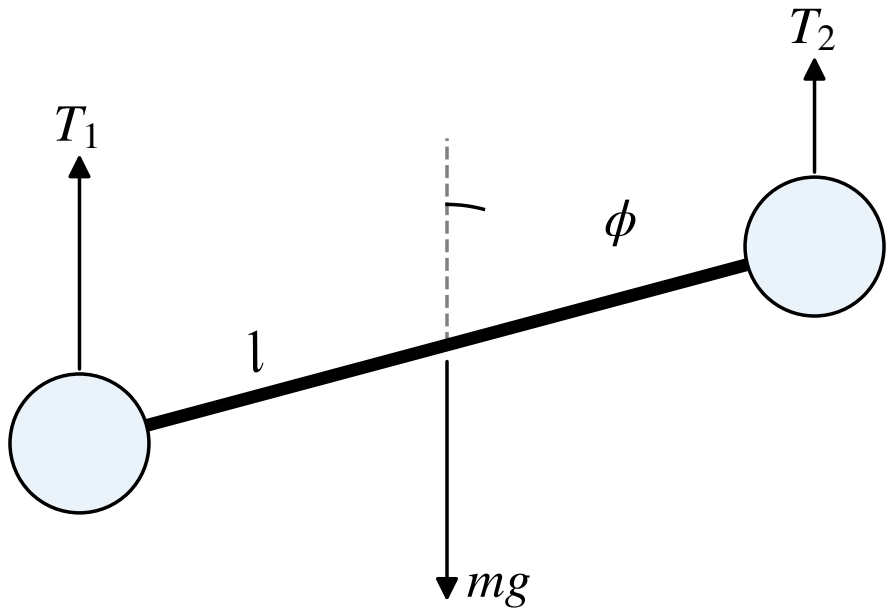

$$\tau_\phi = l(T_1 - T_2)$$

$$J\dot{\omega}_\phi = \tau_\phi$$